

071306T4EEN

ELECTRICAL ENGINEERING (POWER OPTION) LEVEL 6

ENG/OS/PO/CR/10/6

Demonstrate Understanding of Power Generation

March/April 2024



**TVET CURRICULUM DEVELOPMENT, ASSESSMENT AND CERTIFICATION
COUNCIL (TVET CDACC)**

WRITTEN ASSESSMENT GUIDE

INSTRUCTIONS TO ASSESSOR

- i. In this assessment, the candidate will be required to answer **written** questions;
- ii. Marks for each question are indicated in the brackets
- iii. The paper consists of **TWO** sections: **A and B**

This guide consists of TWELVE printed pages.

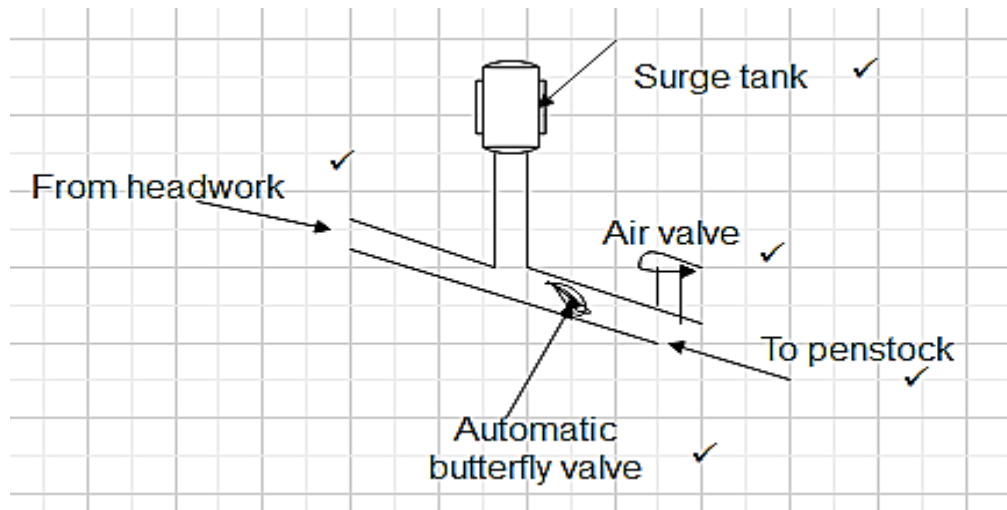
**Assessor should check the guide to ascertain that all the pages are printed as indicated and
that no questions or answers are missing.**

SECTION A (40 MARKS)

1. Outline **four** reasons for using the grid system from power generation station. (4 Marks)
- Stabilizing the power supply i.e., when one generator fails, the other can supply power**
 - Ensure stability in the frequency of the power supply**
 - Use of older plants and less efficient plants to carry peak loads for a short duration**
 - Exchange of peak loads**
 - Reduce plant reserve capacity**
 - Ensure economical operation – arrangement so that more efficient stations work continuously throughout the year at a high load factor.**

(Award 1 or 0 mark for any 4 correct answers)

2. Using a well labelled diagram, describe various penstock protective devices. (8 Marks)



Penstocks are open or closed conduits which carry water to the turbines. Parts include;

- Automatic butterfly valve-shuts off water flow through the penstock promptly if it ruptures.**
- Air valve maintains the air pressure inside the penstock equal to outside atmospheric pressure**
- Surge tank-here water level rises or falls to reduce the pressure swings in the conduit.**

*award 5 marks for correct labelled diagram and 1 mark for each explained point
= total 8 marks*

3. Outline **four** factors considered when selecting site for hydroelectric power generating station. (4 Marks)

- i. **Accessibility**
- ii. **Storage of water**
- iii. **High transmission efficiency**
- iv. **Availability of adequate water**
- v. **Cost and type of land**

(Award 1 mark for any 4 correct response. total = 4 marks)

4. Outline **two** drawbacks of wind as a source of energy. (2 marks)

- a) **Variable output**
- b) **Unreliable because of uncertainty about wind pressure**
- c) **Power generated is quite small**
- d) **Require conversion to a.c**
- e) **Dangerous to some wildlife**
- f) **It is noisy**

(Award 1 mark for any three correct response. Total = 3 marks)

5. Describe the following parts of a nuclear power plant. (4 marks)

- a) Heat exchanger

The coolant gives up heat to the heat exchanger which is utilized in raising the steam from the nuclear reactor.

(Award 2 marks for correct response. Total = 2 marks)

- b) Nuclear reactor

It is an apparatus in which nuclear fuel is subjected to nuclear fission. It controls the chain reaction that starts once the fission is done.

(Award 2 marks for correct response. Total = 2 marks)

6. Outline **four** classifications of generating stations. (4Marks)

- i. **Hydroelectric power stations**
- ii. **Steam power stations**

- iii. **Diesel power stations**
- iv. **Nuclear power stations.**
- v. **Gas turbine power plant.**

(Award 4 marks for correct response. Total = 4 marks)

7. Describe **three** major electrical equipment in any power generating station. (6 marks)
- i. **Alternators – coupled with turbines to convert mechanical energy to electrical energy.**
 - ii. **Transformers -step up the generation voltage for transmission of power. Also used for general service in the power station (lighting).**
 - iii. **Switch gear – it houses equipment which locates the fault on the system and isolate the faulty part from the healthy section.**

(Award 1 mark for stating and 1 mark for description. total = 6 marks)

8. List **TWO** types reaction turbines mainly used in hydro-power plant. (2 Marks)
- i. **Francis turbine**
 - ii. **Kaplan turbine**

(Award 1 Or 0 marks for each correct answer)

9. Explain the following terms as used in power generation. (6 Marks)

a) Kenya Electricity generating company (KenGen)

Manages and develops all public power electricity generating facilities. It sells electricity in bulk to Kenya Power.

b) Independent power producers (IPPs)

Build, own and operate power stations and sell power in bulk to Kenya Power

c) Energy regulatory commission (ERC)

Reviews electricity tariffs and enforces safety and environmental regulations in the power sector as well as safeguarding the interests of electricity consumers.

d) Rural Electricity Authority (REA)

Implements rural electrification projects on behalf of the government.

e) Kenya Electricity Transmission Company (KETRACO)

It is mandated to construct new transmission lines with government funding to accelerate infrastructure development.

f) *Geothermal Development Company (GDC)*

Is tasked with developing steam fields to reduce upstream power development risks so as to promote rapid development of geothermal electric power.

(Award 2 marks for each correct explanation. Total marks = 6 marks)

SECTION B (60 MARKS)

Three questions are tempted (All questions carry equal marks)

10. (a) Explain **three** factors considered when selecting the site for a nuclear power station.

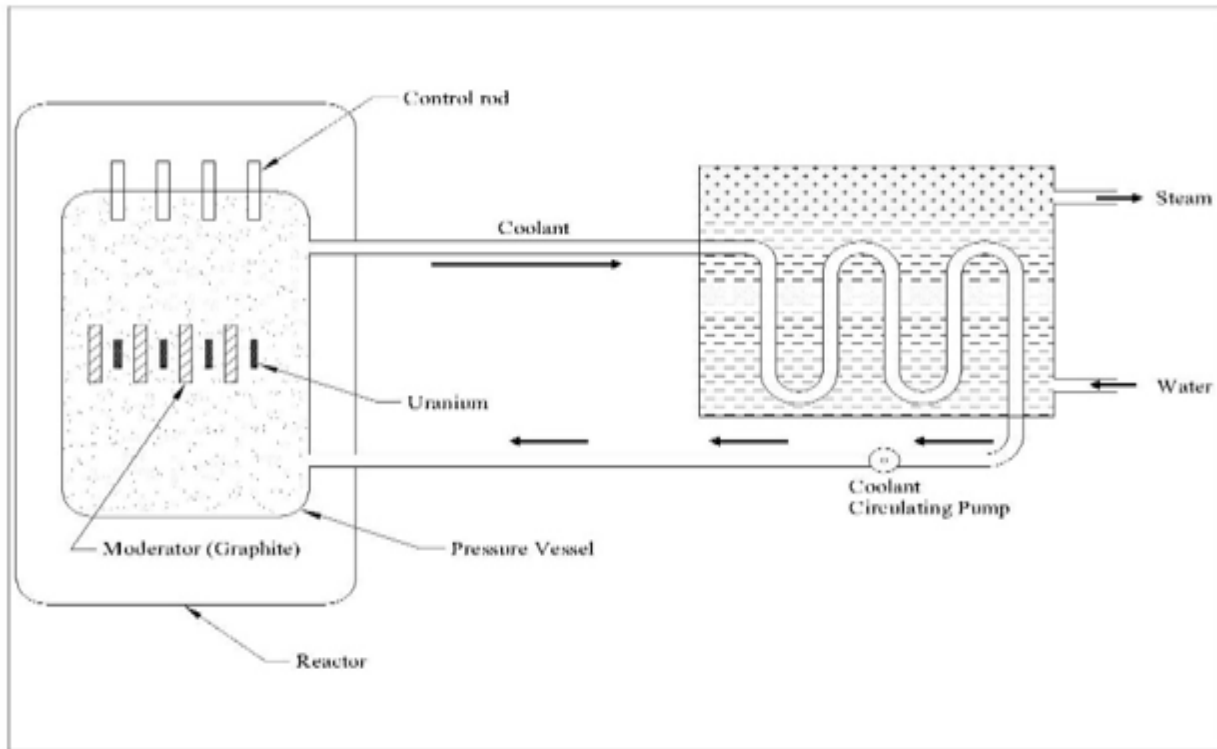
(6 marks)

- a) **Availability of water** – As sufficient water is required for cooling purposes; therefore, the plant site should be located where ample quantity of water is available, *e.g.*, across a river or by sea-side.
- b) **Disposal of waste** - The waste produced by fission in a nuclear power station is generally radioactive which must be disposed off properly to avoid health hazards. The waste should either be buried in a deep trench or disposed off in sea quite away from the sea shore. Therefore, the site selected for such a plant should have adequate arrangement for the disposal of radioactive waste.
- c) **Distance from populated areas**- The site selected for a nuclear power station should be quite away from the populated areas as there is a danger of presence of radioactivity in the atmosphere near the plant. However, as a precautionary measure, *a dome* is used in the plant which does not allow the radioactivity to spread by wind or underground waterways.
- d) **Transportation facilities** - The site selected for a nuclear power station should have adequate facilities in order to transport the heavy equipment during erection and to facilitate the movement of the workers employed in the plant.

(Award 2 marks for each of the three)

b) Draw a labelled diagram of a nuclear reactor.

(8 marks)



(Award 4 marks for correct drawing)

(Award 1 mark for each of the following labelled parts. Total = 4 marks)

- i. **Control rods**
- ii. **Reactor**
- iii. **Uranium**
- iv. **Water & steam**

- c) Water for hydroelectric power station is obtained from a reservoir with a head of 120 meters. If the hydraulic efficiency is 0.86 and electrical efficiency is 0.90; calculate the electrical energy generated per hour per unit cubic meter of water. (6 marks)

Head = 120m

Hydraulic efficiency = 0.86

Electrical efficiency = 0.90

Discharge, $Q = 1\text{m}^3/\text{sec}$ (1 mark)

**$\eta_{\text{overall}} = 0.86 \times 0.90$
= 0.78 (1 mark)**

Wt. of water available/sec

$$W = Q \times 1000 \times 9.81$$

$$= 9810 \quad (1 \text{ mark})$$

$$\text{Power Produced} = W \times H \times \eta_{\text{overall}} \quad (1 \text{ mark})$$

$$= 9810 \times 120 \times 0.78$$

$$= 918216$$

$$= 918 \times 10^3 \text{ watts}$$

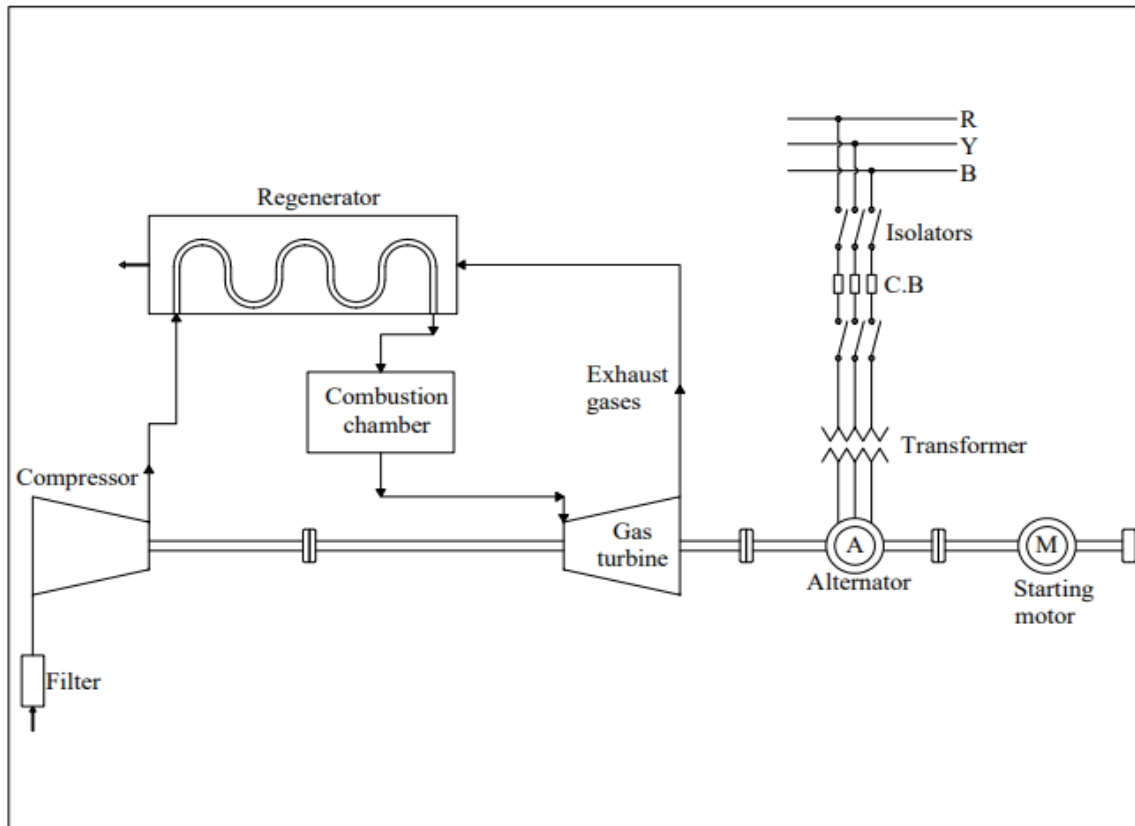
$$= 918 \text{ Kw} \quad (1 \text{ mark})$$

$$\text{Energy generated/hr} = 918 \times 1$$

$$= 918 \text{ kWh}$$

(1 mark)

11. (a) With the aid of a well labelled schematic diagram, describe the operation of a gas turbine power plant. (12 Marks)



(Award 4 marks for correct drawing)

(Award 1 mark for each of any four labelled parts. Total = 4 marks)

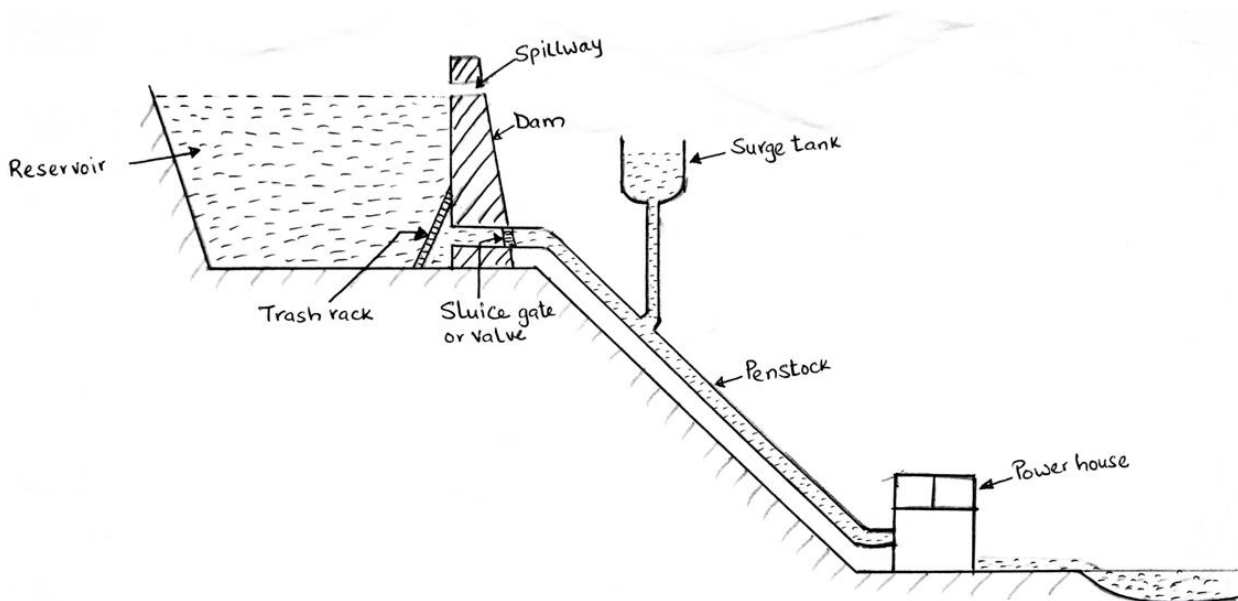
Explanation

- i. In a gas turbine power plant, air is used as the working fluid. The air is compressed by the compressor and is led to the combustion chamber where heat is added to air, thus raising its temperature.
- ii. Heat is added to the compressed air either by burning fuel in the chamber or by the use of air heaters.
- iii. The hot and high-pressure air from the combustion chamber is then passed to the gas turbine where it expands and does the mechanical work.
- iv. The gas turbine drives the alternator which converts mechanical energy into electrical energy.

(Award 4 marks for correct explanation. total = 4 marks)

b) Using a labelled schematic arrangement of hydro-electric power station, explain the function of the following parts. (8 Marks)

- i. Surge tank
- ii. Head works



- i. **Surge tank.** When closed conduits are used always with a surge tank provided. A surge tank which is open at the top reduces the pressure swings in the conduit when water level rises or falls to.

- ii. **Head works.** They are used to discharge the surplus water from the storage reservoir into the river on the down-stream side of the dam. Gates are provided between these piers and surplus water is discharged over the crest of the dam by opening these gates.

(Award 4 marks for correct diagram, Award 2 marks for each correct explanation)

12. (a) An atomic power reactor can deliver 300 MW. If due to fission of each atom of ${}_{92}\text{U}^{235}$, the energy released is 200 MeV, determine the mass of uranium fissioned per hour. (10 marks)

Energy received from the reactor

$$= 300 \text{ MW} = 3 \times 10^8 \text{ W (or Js}^{-1}\text{)}$$

Award 1 mark

$$\text{Energy received/hour} = (3 \times 10^8) \times 3600$$

Award 1 mark

$$= 108 \times 10^{10} \text{ J}$$

Award 1 mark

$$\text{Energy released/fission} = 200 \text{ MeV} = 200 \times 10^6 \times 1.6 \times 10^{-19} \text{ J}$$

Award 1 mark

$$= 3.2 \times 10^{-11} \text{ J}$$

Award 1 mark

Number of atoms fissioned per hour

$$\frac{108 \times 10^{10}}{3.2 \times 10^{-11}} = 33.75 \times 10^{21}$$

Award 2 marks

Now 1 gram-atom (i.e., 235g) has 6.023×10^{23} atoms

Award 2 marks

$\therefore$ Mass of Uranium fissioned per hour

$$\frac{235}{6.023 \times 10^{23}} \times 33.75 \times 10^{21} = 13.17 \text{ g}$$

Award 1 mark

- (b) Explain **five** auxiliary parts of a steam power station.

- i. **Boiler:** this is closed vessel in which water is converted into steam through heating by coal combustion
- ii. **Economizer:** the economizer derives heat from the flue gases to heat feed of water. The feed water is fed to the economizer before heading to the boiler. The economizer extracts a part of heat of flue gases to increase the feed water temperature
- iii. **Super heater:** A super heater is a device which superheats the steam raising the temperature of steam above boiling point of water. This increases the overall efficiency of the plant the

economizer derives heat from the flue gases to heat feed of water. The feed water is fed to the economizer before heading to the boiler.

- iv. **Air preheater:** An air preheater increases the temperature of the air supplied for coal burning by deriving heat from coal gases. Air is drawn from the atmosphere by forced draught fan and passed through air preheater before supplying to the boiler furnace.
- v. **Switch gear:** The equipment at switchgear locates the faulty part on the system and isolates it from the healthy section.

(10 marks)

(Award 2 marks for any 5 correct answer)

- 13. a) Describe the following plant elements of a diesel power station. (8marks)
 - i. Exhaust system. **This system leads the engine exhaust gas outside the building and discharges it into atmosphere. A silencer is usually incorporated in the system to reduce the noise level.**
 - ii. Air intake system. **This system supplies necessary air to the engine for fuel combustion. It consists of pipes for the supply of fresh air to the engine manifold. Filters are provided to remove dust particles from air which may act as abrasive in the engine cylinder.**
 - iii. Cooling system. **This system keeps the temperature of the engine parts within the safe operating limits. The cooling system consists of a water source, pump and cooling towers. The pump circulates water through cylinder and head jacket. The water takes away heat form the engine. The hot water is cooled by cooling towers and is recirculated for cooling.**
 - iv. Engine starting system. **This is an arrangement to rotate the engine initially, while starting, until firing starts and the unit runs with its own power. Small sets are started manually by handles but for larger units, compressed air is used for starting. In the latter case, air at high pressure is admitted to a few of the cylinders, making them to act as reciprocating air motors to turn over the engine shaft. The fuel is admitted to the remaining cylinders which makes the engine to start under its own power**

(Award 4 or 0 for correct answer)

- b) Outline **six** importance of a diesel power plant (6marks)
 - (i) **There are no standby losses.**
 - (ii) **The design and layout is simple**
 - (iii) **It occupies less space**

(iv) It can be located at any place

(v) It requires less quantity of water for cooling

(vi) The overall cost is much less than that of steam power station.

(vii) It requires less operating staff.

(Award 1 or 0 for each correct answer)

(c) Explain the following terms as used in power generation

(6marks)

a) Average load

This refers to the average of loads occurring in a given period.

b) Plant capacity

It is the ratio of actual energy produced to the maximum possible

c) Maximum demand

It is the greatest demand of load on the power station during a given time.

(Award 2 marks for each correct explanation. Total marks = (6marks))

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